🥇 SnowPro Advanced: Data Analyst — Prep Guide

PJ's Academy · Prove you can turn Snowflake data into decisions.

The Data Analyst exam (DAA-C01) validates advanced analytics skills on Snowflake — complex SQL, data prep, visualization, and delivering trustworthy insights.

> **Prerequisite:** SnowPro Core certified.

📋 Exam At A Glance

Item	Detail
Exam code	DAA-C01
Questions	~65
Duration	115 minutes
Cost	$375 USD
Prerequisite	SnowPro Core

Domain Weightings

#	Domain	Weight
1	Data Ingestion & Preparation	25%
2	Data Analysis & Transformation	30%
3	Reporting, Visualization & Delivery	25%
4	Data Quality, Governance & Security	20%

📥 Domain 1 — Ingestion & Preparation (25%)

- Loading data for analysis: `COPY INTO`, stages, external tables, Marketplace datasets.
- Cleaning: dedup (`QUALIFY ROW_NUMBER()`), null handling, type casting, standardizing.
- Reshaping: `PIVOT` / `UNPIVOT`, `FLATTEN` for semi-structured.
- Joining & enriching with Marketplace data (weather, demographics).

📊 Domain 2 — Analysis & Transformation (30%)

Advanced analytical SQL

- Window functions: `LAG` / `LEAD`, running totals, moving averages, `NTILE`, `PERCENT_RANK`.
- `QUALIFY` for top-N and dedup.
- `GROUP BY ROLLUP` / `CUBE` / `GROUPING SETS` for subtotals.
- `MATCH_RECOGNIZE` for funnels & sessionization.
- Recursive CTEs for hierarchies.
- Approximate functions (`APPROX_PERCENTILE`, `APPROX_COUNT_DISTINCT`) for speed.

```
-- Cohort retention with window functions
SELECT cohort_month, months_since,
       COUNT(DISTINCT user_id) AS active,
       ROUND(RATIO_TO_REPORT(COUNT(DISTINCT user_id))
             OVER (PARTITION BY cohort_month) * 100, 1) AS pct
FROM cohorts GROUP BY 1,2;
```

Statistical functions

- `CORR`, `STDDEV`, `VARIANCE`, `REGR_*`, `MEDIAN`, percentiles.

📈 Domain 3 — Reporting, Visualization & Delivery (25%)

- **Streamlit in Snowflake** — build dashboards natively.
- **Snowflake Notebooks** — analysis + narrative.
- BI tools: Tableau, Power BI, Sigma connectivity + best practices.
- Materialized views & result caching for fast dashboards.
- Secure views & shares for delivering data to stakeholders.

🛡️ Domain 4 — Quality, Governance & Security (20%)

- Data quality checks, `GROUP BY` validation, anomaly spotting.
- Masking policies, row access policies (what an analyst sees).
- Understanding RBAC from a consumer's perspective.
- Lineage & trust: ACCESS_HISTORY, freshness checks.

🧠 High-Yield Facts

- **QUALIFY** replaces subqueries for filtering window results — memorize it.
- **ROLLUP/CUBE/GROUPING SETS** produce subtotals & grand totals in one query.
- **MATCH_RECOGNIZE** = funnels/sessionization; know the `PATTERN` / `DEFINE` skeleton.
- **RATIO_TO_REPORT** gives % of group without a self-join.
- **Streamlit in Snowflake** = dashboards with no external hosting.
- **Result cache** (24h) makes repeated dashboard queries free.

📝 Practice Questions (10)

Q1. Which produces subtotals AND a grand total in one query? A) GROUP BY B) GROUP BY ROLLUP C) QUALIFY D) PIVOT

Q2. Best clause to keep the top 3 rows per group inline? A) HAVING B) WHERE C) QUALIFY D) LIMIT

Q3. Which builds funnels/sessionization? A) PIVOT B) MATCH_RECOGNIZE C) FLATTEN D) CUBE

Q4. `RATIO_TO_REPORT` computes: A) A running total B) Value as % of a partition C) A rank D) A median

Q5. Fastest distinct count on billions of rows: A) COUNT(DISTINCT) B) APPROX_COUNT_DISTINCT C) GROUP BY D) DISTINCT

Q6. Native dashboarding inside Snowflake uses: A) Tableau only B) Streamlit in Snowflake C) Excel D) Power BI only

Q7. To remove duplicate rows keeping the latest, use: A) DISTINCT B) QUALIFY ROW_NUMBER() C) GROUP BY D) MERGE

Q8. Which reshapes rows into columns? A) UNPIVOT B) PIVOT C) FLATTEN D) LATERAL

Q9. Repeated identical dashboard queries are free due to: A) Local cache B) Result cache (24h) C) Clustering D) MV

Q10. An analyst sees masked PII because of: A) A view B) A masking policy C) Encryption D) A stage

✅ Answers

1-B · 2-C · 3-B · 4-B · 5-B · 6-B · 7-B · 8-B · 9-B · 10-B

🗓️ 3-Week Plan

- **Week 1:** Advanced analytical SQL (windows, ROLLUP, MATCH_RECOGNIZE).
- **Week 2:** Visualization & delivery (Streamlit, BI, caching).
- **Week 3:** Governance + practice tests.

❄️ *Snowflake Mastery — PJ's Academy*
